Using DDL Statements to Create and Manage Tables

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Naming Rules

- **Table names and column names:**
  - **Must begin with a letter**
  - **Must be 1–30 characters long**
  - **Must contain only A–Z, a–z, 0–9, _, $, and #**
  - **Must not duplicate the name of another object owned by the same user**
  - **Must not be an Oracle server reserved word**

CREATE TABLE Statement

```
CREATE TABLE [schema.]table
        (column datatype [DEFAULT expr][, ...]);
```

- You specify:
  - Table name
  - Column name, column data type, and column size

```
CREATE TABLE dept
        (deptno      NUMBER(2),
        dname        VARCHAR2(14),
        loc          VARCHAR2(13),
        create_date DATE DEFAULT SYSDATE);
```

Including Constraints

- **Constraints enforce rules at the table level.**
- **Constraints prevent the deletion of a table if there are dependencies.**
- **The following constraint types are valid:**
  - NOT NULL
  - UNIQUE
  - PRIMARY KEY
  - FOREIGN KEY
  - CHECK



Defining Constraints

- Column-level constraint:

```
CREATE TABLE employees(
  employee_id  NUMBER(6)
    CONSTRAINT emp_emp_id_pk PRIMARY KEY,
  first_name   VARCHAR2(20),
  ...);
```

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- Table-level constraint:

```
CREATE TABLE employees(
  employee_id  NUMBER(6),
  first_name   VARCHAR2(20),
  ...
  job_id       VARCHAR2(10) NOT NULL,
  CONSTRAINT emp_emp_id_pk
    PRIMARY KEY (EMPLOYEE_ID));
```

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UNIQUE & CHECK Constraints

- Defined at either the table level or the column level:

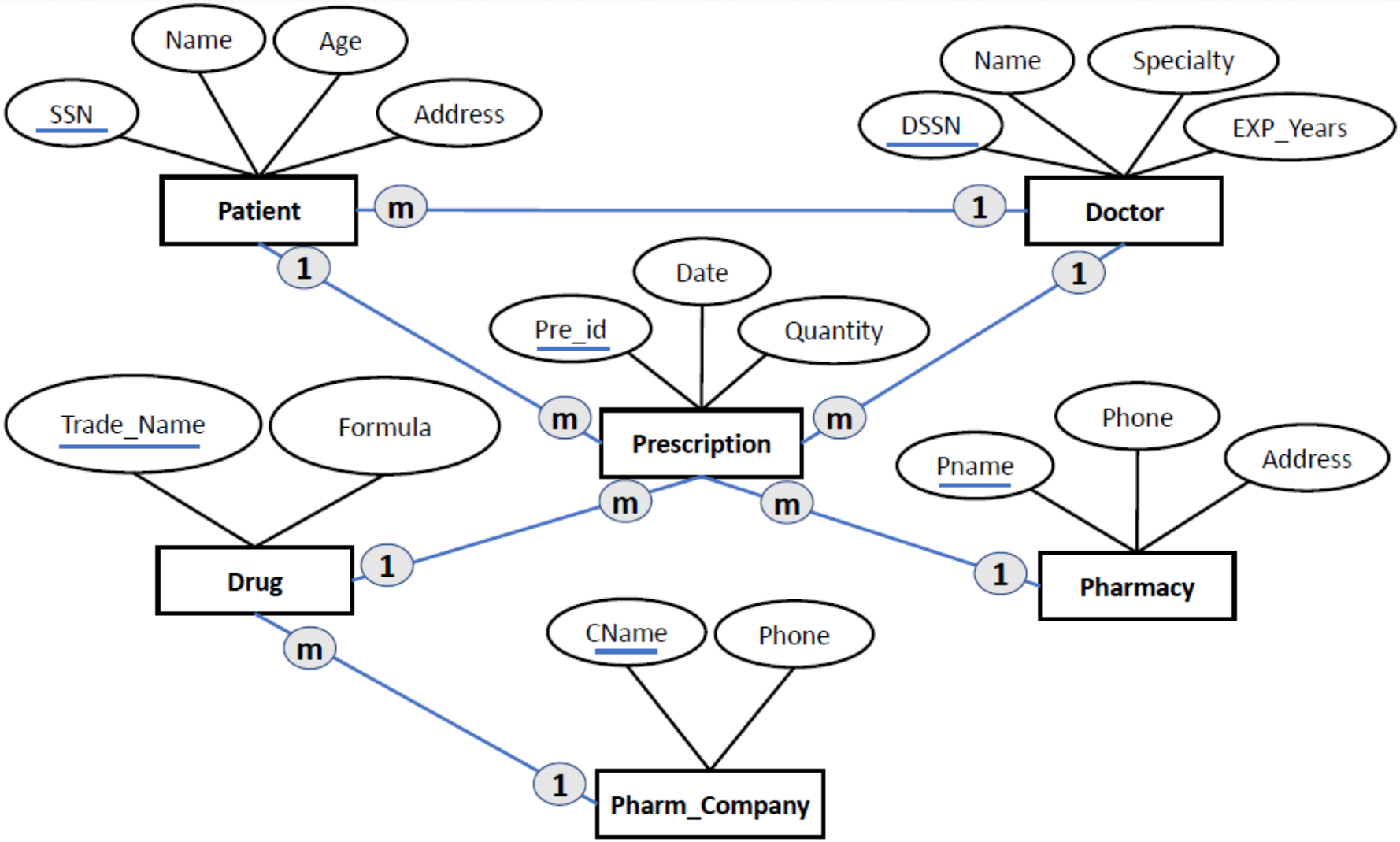
```
CREATE TABLE employees(
    employee_id       NUMBER(6),
    last_name         VARCHAR2(25) NOT NULL,
    email             VARCHAR2(25),
    salary            NUMBER(8,2),
    commission_pct    NUMBER(2,2),
    hire_date         DATE NOT NULL,
...
    CONSTRAINT emp_email_uk UNIQUE(email),
    CONSTRAINT emp_salary_min CHECK (salary > 0)
);
```

FOREIGN KEY Constraint

- Defined at either the table level or the column level:

```
CREATE TABLE employees(
    employee_id       NUMBER(6),
    last_name         VARCHAR2(25) NOT NULL,
    email             VARCHAR2(25),
    salary            NUMBER(8,2),
    commission_pct    NUMBER(2,2),
    hire_date         DATE NOT NULL,
...
    department_id     NUMBER(4),
    CONSTRAINT emp_dept_fk FOREIGN KEY (department_id)
      REFERENCES departments(department_id));
```

PharmacyERD



Task1

1. Write SQL statements to create all the required tables based on the provided Pharmacy ERD .
2. Insert to each table with 50 representative records using appropriate and realistic sample data.

```
CONNECT system/installedpassword;

-- for each section create a new user to build the same schema

CREATE USER Pharmcy IDENTIFIED BY pharmacy;
GRANT DBA TO Pharmcy ;
```

```
CREATE TABLE doctor (
    dssn        NUMBER PRIMARY KEY,
    name        VARCHAR2(100) NOT NULL,
    specialty   VARCHAR2(100),
    exp_years   NUMBER
);
```

```
CREATE TABLE patient (
    ssn       NUMBER PRIMARY KEY,
    name      VARCHAR2(100) NOT NULL,
    age       NUMBER,
    address   VARCHAR2(200),
    dssn      NUMBER,
    CONSTRAINT fk_doctor_patient FOREIGN KEY ( dssn )
       REFERENCES doctor ( dssn )
);
```

```
CREATE TABLE pharmacy (
    pname     VARCHAR2(100) PRIMARY KEY,
    phone     VARCHAR2(20),
    address   VARCHAR2(200)
);
```

```
CREATE TABLE Pharm_Company (
    cname VARCHAR2(100) PRIMARY KEY,
    phone VARCHAR2(20)
);
```

```
CREATE TABLE Drug (
    trade_name VARCHAR2(100),
    formula VARCHAR2(100),
    cname VARCHAR2(100),
    PRIMARY KEY (trade_name, cname),
    CONSTRAINT fk_drug_company FOREIGN KEY (cname)
        REFERENCES Pharm_Company(cname)
);
```

```
CREATE TABLE Prescription (
    pre_id NUMBER PRIMARY KEY,
    pre_date DATE,
    quantity NUMBER,
    patient_ssn NUMBER,
    doctor_dssn NUMBER,
    pharmacy_name VARCHAR2(100),
    drug_trade_name VARCHAR2(100),
    drug_cname VARCHAR2(100),
    CONSTRAINT fk_presc_patient FOREIGN KEY (patient_ssn)
        REFERENCES Patient(ssn),
    CONSTRAINT fk_presc_doctor FOREIGN KEY (doctor_dssn)
        REFERENCES Doctor(dssn),
    CONSTRAINT fk_presc_pharmacy FOREIGN KEY (pharmacy_name)
        REFERENCES Pharmacy(pname),
    CONSTRAINT fk_presc_drug FOREIGN KEY (drug_trade_name, drug_cname)
        REFERENCES Drug(trade_name, cname)
);
```

Illustrate the ALTER statements (Only Illustrate not Apply Code)

- **If I need to add a new column to Doctor table**

```
ALTER TABLE Doctor ADD (email VARCHAR2(100));
```

- **If I need to modify column data type in Patient table**

```
ALTER TABLE Patient MODIFY (address VARCHAR2(300));
```

- **If I need to rename a column in Pharmacy table**

```
ALTER TABLE Pharmacy RENAME COLUMN phone TO phone_number;
```

- **If I need to add a new constraint (check constraint)**

```
ALTER TABLE Patient ADD CONSTRAINT chk_age CHECK (age >= 0 AND age <= 120);
```

- **If I need to add a foreign key from Drug to Pharm_Company (if not already exists)**

```
ALTER TABLE Drug ADD CONSTRAINT fk_drug_company_new
   FOREIGN KEY (cname) REFERENCES Pharm_Company(cname);
```

- **If I need to add a unique constraint**

```
ALTER TABLE Doctor ADD CONSTRAINT uq_doctor_name UNIQUE (name);
```

Illustrate the DROP statements (Only Illustrate not Apply Code)

- **if I need to drop a specific constraint**

```
ALTER TABLE Doctor DROP CONSTRAINT uq_doctor_name;
```

- **if I need to drop a column**

```
ALTER TABLE Doctor DROP COLUMN email;
```

- **if I need to drop tables**

```
DROP TABLE Prescription CASCADE CONSTRAINTS;
DROP TABLE Patient CASCADE CONSTRAINTS;
```

- **if I need to drop the user and all objects**

```
CONNECT system/installedpassword;
DROP USER Section1 CASCADE;
```

Insert 2 records in each table

- INSERT INTO Doctor VALUES (1001, 'Dr. Ahmed Ali', 'Cardiology', 25);
- INSERT INTO Doctor VALUES (1002, 'Dr. Mohamed Hassan', 'Dermatology', 8);
- INSERT INTO Pharm_Company VALUES ('Pharco (1)', '0226374122');
- INSERT INTO Pharm_Company VALUES ('Eva Pharma (2)', '0227707870');
- INSERT INTO Drug VALUES ('Panadol 500mg', 'Paracetamol 500mg', 'Pharco (1)');
- INSERT INTO Drug VALUES ('Augmentin 500mg', 'Omeprazole 20mg', 'Eva Pharma (2)');
- INSERT INTO Pharmacy VALUES ('El-Salam Pharmacy - Branch 1', '01186125617', 'Cairo, Egypt');
- INSERT INTO Pharmacy VALUES ('El-Salam Pharmacy - Branch 2', '01238538251', 'Alexandria, Egypt');
- INSERT INTO Patient VALUES (2001, 'Mohamed Ali', 27, 'Cairo, Egypt', 1001);
- INSERT INTO Patient VALUES (2002, 'Fatma Hassan', 25, 'Alexandria, Egypt', 1002);
- INSERT INTO Prescription VALUES (3001, TO_DATE('2025-09-21','YYYY-MM-DD'), 4, 2001, 1001, 'El-Salam Pharmacy - Branch 1', 'Panadol 500mg', 'Pharco (1)');